

A No-Small-Alias Subcase of the Calderon Sum Formula for Wavelet Systems

agent_lane_02

2026-07-03

Source Problem

The source paper is Ulrik Enstad and Jordy Timo van Velthoven, *On the Calderon sum formula for wavelet systems*, arXiv:2602.10766. Its introduction records the following conjecture of Bownik–Lemvig and Speegle: if an affine wavelet system associated with matrices $A, P \in \text{GL}(d, \mathbb{R})$ and $\psi \in L^2(\mathbb{R}^d)$ is an orthonormal basis, or more generally a Parseval frame, then

$$\sum_{j \in \mathbb{Z}} |\widehat{\psi}((A^t)^j \xi)|^2 = |\det P| \quad \text{for a.e. } \xi \in \mathbb{R}^d.$$

The source proves this under the additional regularity hypothesis $\psi \in \mathcal{B}_\pi$. The result below gives a different subcase which does not assume $\psi \in \mathcal{B}_\pi$.

On the other hand, the aforementioned characterization of Parseval wavelet frames is currently only known for special dilations, such as amplifying dilations [25], dilations expanding on a subspace [21, 22] or dilations satisfying the lattice counting estimate [5]. This has lead to the following conjecture [5, Conjecture 1] and open problem [6, Problem 3.3], which was already implicitly raised in [30, p. 177].¹

Conjecture 1.1 ([5, 6, 30]). *Let $A, P \in \text{GL}(d, \mathbb{R})$ and $\psi \in L^2(\mathbb{R}^d)$. Suppose that*

$$\{|\det(A)|^{-j/2} \psi(A^j \cdot -Pk)\}_{j \in \mathbb{Z}, k \in \mathbb{Z}^d}$$

is an orthonormal basis, or more generally a Parseval frame, for $L^2(\mathbb{R}^d)$. Then the Calderón sum formula holds:

$$\sum_{j \in \mathbb{Z}} |\widehat{\psi}((A^t)^j \xi)|^2 = |\det(P)| \quad \text{for a.e. } \xi \in \mathbb{R}^d. \quad (1.2)$$

¹2020 *Mathematics Subject Classification*. 42C40, 43A80.

Statement

We use the quasi-regular representation notation of Section 4 of the source:

$$\pi(x, A^j)f(t) = |\det A|^{-j/2} f(A^{-j}(t - x)).$$

Thus

$$\pi(\widehat{A^j Pk}, A^j)\psi(\xi) = |\det A|^{j/2} e^{-2\pi i \xi \cdot A^j Pk} \widehat{\psi}((A^t)^j \xi).$$

Theorem 1 (No-small-alias subcase). *Let $A, P \in \text{GL}(d, \mathbb{R})$ and $\psi \in L^2(\mathbb{R}^d)$. Assume*

$$\delta(A, P) := \inf_{\substack{j \in \mathbb{Z} \\ m \in \mathbb{Z}^d \setminus \{0\}}} \|(A^j P)^{-t} m\| > 0.$$

If

$$\mathcal{W}(\psi, A, P) := \{\pi(A^j P k, A^j) \psi : j \in \mathbb{Z}, k \in \mathbb{Z}^d\}$$

is a Parseval frame for $L^2(\mathbb{R}^d)$, then

$$\sum_{j \in \mathbb{Z}} |\widehat{\psi}((A^t)^j \xi)|^2 = |\det P| \quad \text{for a.e. } \xi \in \mathbb{R}^d.$$

More generally, if $\mathcal{W}(\psi, A, P)$ has frame bounds $0 < C_1 \leq C_2 < \infty$, then

$$C_1 |\det P| \leq \sum_{j \in \mathbb{Z}} |\widehat{\psi}((A^t)^j \xi)|^2 \leq C_2 |\det P| \quad \text{for a.e. } \xi \in \mathbb{R}^d.$$

Proof Intuition

The obstruction in the full conjecture is aliasing. At scale j , summing the translation coefficients over $k \in \mathbb{Z}^d$ periodizes in frequency along the dual lattice $(A^j P)^{-t} \mathbb{Z}^d$. If dual lattice vectors from different scales can be arbitrarily small, then no tiny Fourier neighborhood is alias-free at every scale; this is exactly where the source paper needs the coefficient-tail condition $\psi \in \mathcal{B}_\pi$.

Under the no-small-alias hypothesis, we can choose Fourier supports with diameter below $\delta(A, P)$. On such supports every scale has only its diagonal alias. The Parseval frame identity then reads locally as integration against the Calderon sum. Since small cubes cover frequency space, the local identity forces the Calderon sum formula almost everywhere.

Proof

We first record the elementary lattice calculation used at each scale.

Lemma 1 (Alias-free lattice summation). *Let $B \in \text{GL}(d, \mathbb{R})$, let $E \subset \mathbb{R}^d$ be measurable, and assume*

$$(E - E) \cap (B^{-t} \mathbb{Z}^d \setminus \{0\}) = \emptyset.$$

If $G \in L^2(\mathbb{R}^d)$ is supported in E , then

$$\sum_{k \in \mathbb{Z}^d} \left| \int_{\mathbb{R}^d} G(\xi) e^{2\pi i \xi \cdot Bk} d\xi \right|^2 = |\det B|^{-1} \int_E |G(\xi)|^2 d\xi.$$

Proof. Let Q be a measurable fundamental domain for the dual lattice $B^{-t} \mathbb{Z}^d$. Periodize

$$G_B(\eta) = \sum_{\ell \in B^{-t} \mathbb{Z}^d} G(\eta + \ell), \quad \eta \in Q.$$

The assumption on $E - E$ implies that at most one summand is nonzero for a.e. $\eta \in Q$, hence

$$\int_Q |G_B(\eta)|^2 d\eta = \int_E |G(\xi)|^2 d\xi.$$

Moreover

$$\int_{\mathbb{R}^d} G(\xi) e^{2\pi i \xi \cdot Bk} d\xi = \int_Q G_B(\eta) e^{2\pi i \eta \cdot Bk} d\eta.$$

The functions $|\det B|^{1/2} e^{2\pi i \eta \cdot Bk}$, $k \in \mathbb{Z}^d$, form an orthonormal basis of $L^2(Q)$. Parseval's identity on Q gives the claim. $\square$

We now prove the theorem. Put

$$h_j(\xi) := |\det A|^{j/2} \widehat{\psi}((A^t)^j \xi), \quad B_j := A^j P.$$

Then

$$\pi(\widehat{A^j P k}, A^j) \psi(\xi) = e^{-2\pi i \xi \cdot B_j k} h_j(\xi).$$

Let $E \subset \mathbb{R}^d$ have diameter strictly smaller than $\delta(A, P)$, and let $f \in L^2(\mathbb{R}^d)$ be such that $\widehat{f}$ is bounded and supported in E . For every j , the no-small-alias condition gives

$$(E - E) \cap (B_j^{-t} \mathbb{Z}^d \setminus \{0\}) = \emptyset.$$

Applying Lemma 1 with $G(\xi) = \widehat{f}(\xi) \overline{h_j(\xi)}$ yields

$$\begin{aligned} \sum_{k \in \mathbb{Z}^d} |\langle f, \pi(A^j P k, A^j) \psi \rangle|^2 &= |\det B_j|^{-1} \int_E |\widehat{f}(\xi)|^2 |h_j(\xi)|^2 d\xi \\ &= |\det P|^{-1} \int_E |\widehat{f}(\xi)|^2 |\widehat{\psi}((A^t)^j \xi)|^2 d\xi. \end{aligned}$$

Summing over j and using Tonelli's theorem, the frame inequalities for $\mathcal{W}(\psi, A, P)$ give

$$C_1 \int_E |\widehat{f}(\xi)|^2 d\xi \leq |\det P|^{-1} \int_E |\widehat{f}(\xi)|^2 \sum_{j \in \mathbb{Z}} |\widehat{\psi}((A^t)^j \xi)|^2 d\xi \leq C_2 \int_E |\widehat{f}(\xi)|^2 d\xi.$$

Since this holds for all bounded $\widehat{f}$ supported in E , it follows that

$$C_1 |\det P| \leq \sum_{j \in \mathbb{Z}} |\widehat{\psi}((A^t)^j \xi)|^2 \leq C_2 |\det P| \quad \text{for a.e. } \xi \in E.$$

Finally, $\mathbb{R}^d$ is a countable union of cubes of diameter less than $\delta(A, P)$. Applying the preceding conclusion to each cube proves the global a.e. statement. In the Parseval case $C_1 = C_2 = 1$, this is precisely the Calderon sum formula. $\square$

Examples And Scope

The condition $\delta(A, P) > 0$ is an arithmetic separation condition on the reciprocal translation lattices. It holds, for example, whenever the lattices $(A^j P)^{-t} \mathbb{Z}^d$ range over a uniformly discrete family. A concrete class is given by lattice automorphism regimes: after conjugating by P , if A^{-t} acts as an automorphism of a fixed lattice, then all reciprocal lattices are the same lattice and $\delta(A, P) > 0$.

The result does not cover the usual expansive case. If A is expansive, then nonzero vectors of $(A^j P)^{-t} \mathbb{Z}^d$ typically approach zero as $j \rightarrow +\infty$, so the alias-free localization argument cannot start. That is consistent with the source paper's use of a different mechanism, namely the coefficient-space condition $\psi \in \mathcal{B}_\pi$.

Novelty And Verification Notes

Local run indexes were searched for arXiv:2602.10766 and for Calderon/wavelet keywords; no existing exact packet was found. Web searches for the exact Calderon-sum conjecture and Bownik–Lemvig problem located the source arXiv paper but no later resolution. This packet claims only the no-small-alias subcase, not the full conjecture.

The proof uses only Plancherel/Parseval on the dual lattice and the frame identity. No computational verification is involved.

References

1. U. Enstad and J. T. van Velthoven, *On the Calderon sum formula for wavelet systems*, arXiv:2602.10766.
2. M. Bownik and J. Lemvig, references cited in the source as the origin of the Calderon-sum conjecture/open problem.
3. D. Speegle, reference cited in the source as an earlier implicit formulation of the same problem.

Human Review Recommendation

Recommended review focus: check the normalization convention between the source’s displayed affine system and its Section 4 representation notation, and verify the determinant factor in Lemma 1. The mathematical mechanism is otherwise compact: the no-small-alias condition removes every off-diagonal alias term.